

CRP 1200: Introduction to Statistics for Urban Studies Spring 2025

Course time and location: MW, 8:40am - 9:55am at Sibley 305

Instructor: Wenzheng Li, Ph.D.

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My Office Hours: Monday 2:30 – 4:30pm and Wednesday 11:00am-1:00pm in Sibley Hall 214. Book a time [here](#).

In addition, Sebastian Restrepo Rodriguez (sr2323@cornell.edu) will serve as the teaching assistant and Jinpeng Yang (jy978@cornell.edu) and Yuchen Xie (yx646@cornell.edu) will serve as GTRs for this course. They will hold office hours each week in addition to my office hours in Sibley 305. The times for their office hours will be posted on Canvas.

Course Description: The course is an introduction to descriptive and inferential statistics, which include graphical tools, numerical measures, univariate and multivariate methodologies, to better understand places and the well-being of people. You will learn how to use statistics to illuminate your understanding of the world, analyze and evaluate policy and predict. You will learn how to empirically test theories about social phenomena. You will use excel, STATA and be introduced to R. The course will expose you to important datasets used in urban studies contexts.

Learning Objectives:

1. Learn empirical methods that help examine and explain important social phenomena. This includes learning how to use numeric and graphical tools to describe data (descriptive statistics).
2. Learn to apply and analyze social science reasoning through empirical and graphical methods. In particular, learn how to use statistical inference to test theories and address research questions (inferential statistics).
3. Learn how to use graphical and written modalities to effectively communicate your statistical findings.

Comparable Courses:

The material in this course is comparable to the content in AEM 2100, ENGRD 2700, HADM 2010, ILRST 2100, MATH 1710, PAM 2100, PAM 2101, PSYCH 2500, SOC 3010, STSCI 2100, STSCI 2150. This course can count as the prerequisite for any course that requires one of these as a prerequisite.

Textbook:

Intro Stats, 6th edition by Richard De Veaux, Paul Velleman and David Bock, Pearson Publishing, 2022.

This book is available through instant access on the course canvas site.

Optional readings: There is a lot of recent public interest in the role statistics and data analysis can play in public policy and urban studies. Several good popular books have been written that you might enjoy and will likely enhance your learning. Here are a few:

1. Ayers, Ian (2008), *Super Crunchers: Why Thinking by the Numbers is the New Way to be Smart*.
2. Kahneman, Daniel (2011), *Thinking, Fast and Slow*.
3. Kahneman, D., O. Sibony, and C. Sunstein (2021) *Noise: A Flaw in Human Judgement*
4. Silver, Nate (2012), *The Signal and the Noise: Why so many Predictions Fail*.
5. Wheelan, Charles (2013), *Naked Statistics: Stripping the Dread from the Data*.

Software and Software manuals: We will be using both STATA and Excel this semester. STATA is available for you to access remotely through Cornell's "apps on demand". Instructions will be provided on how to access the canvas site where the apps will be available. STATA is also available in the computer classrooms on campus. Besides Sibley 305, the Barclay Jones Lab, STATA is installed in the computer labs in Mann Library, Management Library, Olin & Uris Libraries, and Phillips Hall and other locations.

Attendance: Attendance is very important and required in this course. If you cannot be in class because you are unwell, please let me know. Attendance will be taken.

Course Evaluation:

1. Attendance and in-class participation (20%) Students will be expected to actively engage in class. There will be a lot of in-class problem solving and exercises. Students may be asked to submit work they have done in class for evaluation. Students who miss four or more classes will automatically lose 5% of this attendance grade.

2. Problem Sets (50%): There will be approximately 5 or 6 problem sets this semester. They will be assigned roughly every other week. You will always have at least a week to complete the assignment. Late Problem Sets will automatically be downgraded unless there is a medical or family emergency:

2%: if turned in the following day

8%: if turned in within 1 week

15%: after 1 week

3. Final Exam (30%): There will be a take-home final exam. Since the exam is taken at home, please be sure you fully understand the code of academic integrity with regard to take-home exams. You are especially obliged to understand that "between the time a take-home exam is distributed and the time it is submitted by the student for grading, the student may not consult with any persons other than the course professor regarding the exam."

Inclusivity Statement

We understand that our members represent a rich variety of backgrounds and perspectives. The Department of City and Regional Planning is committed to providing an atmosphere for learning that respects diversity. While working together to build this community we ask all members to:

- share their unique experiences, values, and beliefs.
- be open to the views of others.

- honor the uniqueness of their colleagues.
- appreciate the opportunity that we have to learn from each other in this community.
- value each other's opinions and communicate in a respectful manner.
- keep confidential discussions that the community has of a personal (or professional) nature.
- use this opportunity together to discuss ways in which we can create an inclusive environment in this course and across the Cornell community.

Bias Reporting

If you have seen, heard, or experienced bias in any form you should make a report. This helps to track our culture, progress, and flag problematic behavior and can help in identifying patterns long term. A bias report is not a punitive action, and can be made anonymously.

Since 2000, Cornell University has had a program to track bias that is occurring on all campuses in an effort to be proactive in creating an inclusive climate for all. You can access resources or report incidents that are experienced or witnessed, in campus, university event or virtually using the Bias Reporting Portal at: <https://diversity.cornell.edu/our-commitments/bias-reporting-cornell>

In addition to university-wide civil rights compliance, the Office of Institutional Equity and Title IX (OIETIX) is responsible for collecting and tracking all reported bias activity that occurs at Cornell University that could potentially impact our commitment to diversity and inclusion, including all reports made by faculty, staff, students, and visitors to the Ithaca, Weill Cornell Medicine, and Cornell Tech campuses.

Office of Institutional Equity & Title IX

Students who experience sexual violence, sexual harassment, or discrimination based on gender or sexual identity are encouraged to report their experience to the Office of Institutional Equity and Title IX titleix@cornell.edu to explore formal and informal reporting options, and explore the support and resources available. Information shared in class assignments, class discussions, and at public events do not constitute an official disclosure, and faculty and staff do not have to report these to the Title IX Coordinator. Any other disclosure to faculty and staff needs to be reported to the Title IX Coordinator.

Nearly all faculty and staff members are legally required to disclose any reports of sexual misconduct to the Title IX Coordinator and cannot guarantee confidentiality. Confidential resources are not required to disclose reports of sexual misconduct to the

Title IX Office. All students have the right to report to local law enforcement (Cornell Police, Tompkins County Sheriff, or New York State Police.) All students have the right to be protected by the University from retaliation for reporting an incident and to receive assistance and resources from the University. More information at: <https://titleix.cornell.edu>

Religious Observances

Cornell University is committed to supporting students who wish to practice their religious beliefs. Students are urged to discuss religious absences with their instructors well in advance of

the religious holiday so that arrangements for making up work can be resolved before the absence. Cornell's faculty are governed by the Faculty Handbook, which requires them to provide reasonable accommodations to students when their religious observance conflicts with exam-taking, class attendance, and other course-related requirements in compliance with New York State law. The Office of Spirituality and Meaning-Making (OSMM) maintains a religious accommodation website that is an extremely valuable resource for both students and faculty. If you have questions or concerns, you may call the OSMM office at 607.255.6002 or email Oliver Goodrich, Associate Dean of Students for Spirituality and Meaning-Making. You may also contact the Office of the Dean of the Faculty by email or by phone at 607.255.4843.

Students with Disabilities

Your access in this course is important to me. Please request your accommodation letter early in the semester, or as soon as you become registered with SDS, so that we have adequate time to arrange your approved academic accommodations.

Once SDS approves your accommodation letter, it will be emailed to both you and me. Please follow up with me and the Phd TA to discuss the necessary logistics of your accommodations.

- If you are approved for exam accommodations, please consult me and the Phd TA at least two weeks before the scheduled exam date to confirm the testing arrangements
- If you experience any access barriers in this course, such as with printed content, graphics, online materials, or any communication barriers; reach out to me or SDS right away.
- If you need an immediate accommodation, please speak with me after class or send an email message to me and SDS at sds_cu@cornell.edu.
- If you have, or think you may have a disability, please contact Student Disability Services for a confidential discussion: sds_cu@cornell.edu, 607-254-4545, sds.cornell.edu. More information:

Code of Academic Integrity

Each student in this course is expected to abide by the Cornell University Code of Academic Integrity. Any work submitted by a student in this course for academic credit will be the student's own work, except in the case of certain collaborative projects as designated by the professor. The full text of the Cornell Code of academic integrity may be found at

<https://cuinfo.cornell.edu/aic.cfm>

Syllabus Change Policy

Except for changes that substantially affect implementation of the evaluation (grading), the activities and assignments described below will be subject to change at any time during the semester to allow class time and activities to maximize the course's benefit to the students, depending on their ongoing development and interests.

Documentation

Course work and activities may be captured and recorded to document course activities and/or progress. The documentation and display may include identifiable student images, voices, and likenesses.

Course Schedule:**Section 1: Intro to Stats and Descriptive Analysis**

January 22-27: Introduction to Data and Statistical Concepts

Chapter 1

January 27-February 5: Univariate descriptive statistics: displaying and describing data

Chapters 2 and 5 (chapter 4 is optional)

February 10-19 (*February break: Feb 15-18*): Bivariate descriptive statistics with quantitative data, including linear regression

Chapters 6 and 7

February 24: Bivariate descriptive statistics with categorical data

Chapter 3

Section 2: Inferential Statistics and Probability Distribution

February 26-March 10: Introduction to Inferential Statistics, Sampling and Probability

Chapter 10 and 12

March 12-24: Sampling Distributions, Central Limit Theorem and Confidence Intervals

Chapter 13 and 14

March 24-April 14 (*Spring Break: March 29 – April 6*): Testing Hypotheses

Chapter 15 and 16

April 16-21: Comparing Groups

Chapter 17

Section 3: Fundamentals of Statistical Modeling

April 23-28: Multiple Regression

Chapter 9

April 30-May 5: The Assumptions of a Multiple Regression Model, Model Specification Issues and Inference

Chapter 8 and 20